|INCH-POUND|
MIL-G-25507B(AS)
15 April 1991
SUPERSEDING
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MILITARY SPECIFICATION

GAGE, PRESSURE, DIAL INDICATION, OXYGEN 0-2000 PSI, TYPE MF-2

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 1 1/2 inch round dial, high pressure oxygen system pressure gage (range 0-2000 psi).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2b).

SPECIFICATIONS

FEDERAL

BB-A-1034	-	Air, Compressed, for Breathing Purposes.
BB-N-411	-	Nitrogen, Technical.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6685
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-P-116 - Preservation, Methods of

MIL-P-7105 - Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for

STANDARDS

FEDERAL

FED-STD-H28/2 - Unified Inch Screw Threads, UN and UNR Thread Forms

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S Military Property

MIL-STD-889 - Dissimilar Metals

MIL-STD-1359 - Cleaning Methods and Procedures for Breathing Oxygen Equipment

MS33558 - Numerals and Letters, Aircraft Instrument Dial Standard Form of

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2b).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B40.1-85 - Gauges, Pressure and Vacuum, Indicating Dial Type-Elastic Element

(Application for copies should be addressed to the American National Standards, Institute, 1430 Broadway, New York, NY 10018-3308.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2c), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to oxygen, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.2.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.3 Nonferrous. Nonferrous materials shall be used for all parts of the gage except where ferrous materials are essential and do not come in direct contact with the oxygen.

3.2.4 Nonmagnetic. Nonmagnetic materials shall be used for all parts of the gage except where magnetic materials are essential.

3.2.5 Protective treatment. When materials are used in the construction of the gage that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.3 Standard parts. MS and AN standard parts shall be used where they suit the purpose.

3.4 ANSI standard. The gage shall conform to ANSI B40.1-85.

3.5 Design and construction. The gage shall be designed and constructed so that no part will work loose in service. They shall be leakproof, and shall be built to withstand without failure the strains, jars, vibrations, and other conditions incident to shipping, storage, installations, and service. Any pivots, bearings, gears or other moving parts shall neither bind nor shake, and shall be as near frictionless as practicable. The gages shall be so designed that minor repairs can be easily made by the personnel of the operating units without the aid of special tools.

3.5.1 Case. The case shall be in accordance with Figure 1. The case shall be made of either nonferrous low-density metal or a synthetic material and shall have a smooth surface with a durable dull black finish. The synthetic material shall be of the thermal-setting type. The case shall be of a durable construction to prevent damage to the gage mechanism under conditions incident to shipping, storage, installation and service.

3.5.2 Bosses. The bosses shall be as shown in figure 1. Wrench flats shall be provided on the sides of the boss.

3.5.3 Operating mechanism. The indicating and the pressure elements shall comprise a unit distinct from the case. This unit shall be sufficiently rugged in order that small distorations of the case will not affect the accuracy of the indications.

3.5.3.1 Gage adjustment. The gage shall be provided with a means to adjust or correct the indications during the service life of the gage. Adjustment shall require no special tools.

3.5.4 Dial. The dial shall be held securely in such a manner that it will not loosen or turn when the gage is vibrated. If screws are used to attach the dial plate, they shall be so located that they will not interfere with any dial marking or prevent locating any dial marking in the proper location.

3.5.4.1 Dial face. The dial face shall be as shown in figure 2.

3.5.5 Pointer. The pointer shall be in accordance with figure 3. It shall be light and sufficient rigid to prevent oscillation under vibration, and shall be free to swing beyond either end of the range without being engaged by a mechanical stop. Suitable stops shall be incorporated within the mechanism to prevent disengagement if possible due to operating mechanism design. The pointer shall be of such length as to touch but never overlap the graduations on the dial face. The indicating end of the pointer shall be painted white.

3.5.5.1 Overpressure stop. An overpressure stop shall be provided on the gage, designed in such a manner as to restrain the mechanism rather that the pointer. The overpressure stop shall allow pointer travel beyond full scale, 2100 ± 5 psi.

3.5.5.2 Hair springs. Hair springs, if used, shall be made from phosphor bronze.

3.5.6 Surge pressure restriction. A restriction of 0.020 inch diameter shall be provided between the gage connection and the gage mechanism.

3.6 Performance.

3.6.1 Visual examination and dimensional check.

3.6.1.1 Visual examination. When visually examined as specified in 4.6.1.1, the gage shall conform to the requirements of this specification. Table IV, Classification of Defects for Visual Examination, shall be used to classify and enumerate the defects.

3.6.1.2 Dimensional check. When dimensionally checked as specified in 4.6.1.2, the gage shall conform to the requirements of figures 1, 2, and 3. Any measurement deviating from the specified dimensions and tolerances shall be classified as a dimensional defect.

3.6.2 Scale error (room temperature). The gage when tested as specified in 4.6.2, shall read within the scale tolerance specified in Table I.

3.6.3 Friction. The friction error of the gage, when tested as specified in 4.6.3, shall not exceed 50 psi.

3.6.4 Leakage. The pressure readings of the gage, when tested as specified in 4.6.4, shall be the same.

3.6.5 Low temperature and cold resistance. The gage when tested as specified in 4.6.5, shall show no evidence of material failure and the gage pointer shall move smoothly throughout the entire range of travel. The pressure readings for the scale error and friction test points shall be within the tolerance specified in Table I and in 3.6.3.

3.6.6 High temperature. The gage, when tested as specified in 4.6.6, shall not show any evidence of material failure and the pressure readings for the scale error test points shall be as specified in Table I.

3.6.7 Vibration. The gage, when tested as specified in 4.6.7, shall not show any evidence of material failure and the pressure readings for the scale error test points shall be as specified in Table I.

3.6.8 Seasoning. The gage, when tested as specified in 4.6.8, shall not show any evidence of material failure and the gage pressure readings for the scale error (room temperature) test points shall be as specified in Table I.

3.6.9 Drift. The gage pressure reading, when tested as specified in 4.6.9, shall not increase more than 30 psi.

3.6.10 Overpressure. The gage, when tested as specified in 4.6.10, shall show no evidence of material failure and the pressure readings for the scale error (room temperature) test points shall be as specified in Table I.

3.7 Threads.

3.7.1 Screw threads. Screw threads shall conform to FED-STD-H28/2.

3.7.2 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Weight. The weight of the gage shall not exceed 0.20 pounds.

3.10 Identification marking. The gage shall be marked in accordance with MIL-STD-130 and shall contain the following information:

GAGE, PRESSURE, DIAL INDICATING, OXYGEN
0-2000 PSI, Type MF-2
MIL-G-25507
MANUFACTURER'S IDENTIFICATION
CONTRACT OR ORDER NO.

3.11 Cleanliness. The gage shall be free of oil, grease and other organic material.

3.12 Workmanship. The gage shall be uniform in quality and shall be free from irregularities, defects or foreign matter which could adversely affect safety, performance, reliability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. First article inspection consists of tests and examinations performed on samples which are representative of the production item after award of contract to determine that the production item conforms to the requirements of this specification (see 4.3).

b. Quality conformance inspection. Quality conformance inspection consists of tests and examinations performed on individual products or lots with the requirements set forth in this specification (see 4.4).

4.3 First article inspection. First article inspection shall consist of all the examinations and tests specified in Table II.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit three complete assembled gage samples. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a manufacturer is in continuous production of the gages from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2d). Approval of the first article inspection sample or waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2e).

4.3.1.1 First article report. Upon completion of the first article inspection program, the Government activity responsible for conducting the program (see 6.2e) shall report to the Contracting Officer the results of the program, with appropriate recommendations.

4.3.1.2 First article sample disposition. The samples will be destroyed during the inspection program and will not be returned to the contractor.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table III. The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-105.

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Gages. An inspection lot shall be expressed in units of one gage, made under essentially the same conditions and from the same materials. The sample unit shall be one gage.

4.4.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing gages, fully prepared for delivery, made from the same materials and components. The sample unit shall be one shipping container, containing gages, fully prepared for delivery with the exception that it need not be sealed.

4.5 Test conditions.

4.5.1 Temperature and pressure. Unless otherwise specified herein, tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. The temperature and barometric pressure shall be recorded at the time of inspection and, when required, the test results shall be corrected to normal temperature and pressure (NTP) conditions. NTP conditions are 70oF and 29.92 inches of mercury.

4.5.2 Test gas. Unless otherwise specified, water pumped oil free nitrogen conforming to BB-N-411 or compressed breathing air conforming to BB-A-1034 shall be used in testing of the gages.

4.5.3 Tapping. Unless otherwise specified, the gage shall be tested in position and shall be lightly tapped before a test reading is taken.

5.2.2 Level B. Each unit, packaged as specified in 5.1, shall be packed for shipment in containers conforming to PPP-B-601 or PPP-B-621, Domestic Type; fiber-board boxes conforming to PPP-B-636 or PPP-B-640, Weather Resistant may be used provided that the gross weight is not greater than the applicable specification limitation.

5.2.3 Level C. Units, packaged as specified in 5.1, shall be packed for shipment in exterior-type shipping containers conforming to rules and regulations pertaining to mode of transportation and acceptable to carrier at the lowest possible tariff rate that will ensure safe delivery. Contractor's commercial/ industrial packing procedure may be used when the above criteria are met.

5.3 Marking. The exterior and interior containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall be required on each unit package and shipping container:

CAUTION

DO NOT ALLOW PETROLEUM CONTAMINANTS OF ANY KIND TO BE USED/STORED ON OR ABOUT THESE CONTAINERS.

5.4 Special requirements. All wrappings, cushioning, dunnage and containers used in preservation, packaging and packing of regulators shall be completely free of contamination by oil or grease.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pressure gage covered by this specification is intended for use in indicating oxygen supply pressure in high pressure aircraft oxygen systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification, including any amendments.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. When first article is required (see 3.1).

d. Whether first article inspection is waived (see 4.3.1).

e. Name and address of the first article inspection laboratory (see 4.3.1) and the name of the Government activity responsible for conducting the first article inspection program (see 4.3.1.1).

f. Applicable levels of preservation, packaging, and packing (see 5); including marking requirements (see 5.3).

g. Items of data required (see 6.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3.1.1	DI-T-5329	Inspection test reports	Use contractor format

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Aircraft oxygen systems
Bourdon tube
Compressed gas system
Dial indicating
Oxygen
Oxygen breathing regulator
Pressure gage
Pressure sensing

Preparing Activity
Navy-AS
Project No. 6685-N844

TABLE I. Test requirements.

Test Pressure (psi)	Tolerance (psi) Room Temperature	Tolerance (psi) Low Temperature, High Temp, Vibration, Overpressure
0	+50	+75
200	+50	+75
400	+50	+75
600	+50	+75
800	+50	+75
1000	+50	+75
1200	+50	+75
1400	+50	+75
1600	+50	+75
1800	+50	+75
2000	+50	+75

TABLE II. First article inspection.

Inspection	Requirement	Test method
Visual & dimensional examination	3.6.1	4.6.1
Scale error (room temperature)	3.6.2	4.6.2
Friction	3.6.3	4.6.3
Leakage	3.6.4	4.6.4
Low Temperature & cold resistance	3.6.5	4.6.5
High temperature	3.6.6	4.6.6
Vibration	3.6.7	4.6.7
Seasoning	3.6.8	4.6.8
Drift	3.6.9	4.6.9
Overpressure	3.6.10	4.6.10

TABLE III. Quality conformance inspection.

Inspection	Method	Sample size	Acceptance criteria
Visual and dimensional examination	4.6.1.1 and 4.6.1.2	Every gage for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defect. An Acceptable Quality Level of 2.5 defects per hundred units for minor defects.
Scale error (room temp)	4.6.2	Every gage	Reject all defective units
Friction	4.6.3	Every gage	Reject all defective units
Vibration	4.6.7	Inspection level III 1/	Acceptance number zero, rejection number 1
Seasoning	4.6.8	Inspection level III 1/	Acceptance number zero, rejection number 1
Drift	4.6.9	Inspection level III 1/	Acceptance number zero, rejection number 1
Overpressure	4.6.10	Inspection level III 1/	Acceptance number zero, rejection number 1
Packaging	4.6.1.3	Inspection level II 1/	An acceptable quality level of 2.5 defects per 100 units.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.